Accessibility & performance lab

Seven completed stages of the CineReview movie-review site.

Performance: 76 to 100/100
Accessibility: 70 to 100/100

The baseline is retained, and each stage is a separate cumulative page. The final solution adds working search, keyboard-operable detail buttons and accessible newsletter validation. Open the lab guide.

Step	Completed change
1	Semantic landmarks, structured headings and skip navigation.
2	Real buttons, working search/details and descriptive destination links.
3	Accurate image alternatives; empty alt for a decorative repeated image.
4	Five text contrast failures corrected to at least 4.5:1.
5	Visible labels, associated errors, invalid state and error focus.
6	Dimensions for every image; lazy loading for the lower featured image.
7	External classic scripts at body end with defer.

Method and limits

Lighthouse 13.6.0; mobile emulation, simulated network/CPU throttling; one before and one after run on a local server. The final page uses local copies of the same placeholder photographs, replacing remote image redirects. That origin change contributes to the combined result. Scores can vary, and an automated Accessibility score of 100 does not establish full WCAG conformance.

Browser validation

Checked title search using Enter; details using Enter and Space; aria-expanded state; missing-name and invalid-email errors; focus on the first invalid input; correction and valid local submission; image dimensions; deferred scripts; and 390px mobile reflow. No page errors occurred during those interaction checks.

Steps 1-3 / Structure and controls

Make the page understandable and operable without a mouse.

1 / Semantic HTML

header, nav, main, section, article and footer identify the structure. The movie list contains individual articles, and headings describe the sections. A first-in-body skip link lets keyboard users bypass repeated navigation.

```
<a class="skip" href="#main">Skip to main content</a>
<header><nav aria-label="CineReview">...</nav></header>
<main id="main" tabindex="-1">
  <section aria-labelledby="now-showing">
    <h2 id="now-showing">Now Showing</h2>
    <ul class="movie-grid"><li><article>...</article></li></ul>
  </section>
</main>
<footer>...</footer>
```

2 / Buttons and meaningful links

Clickable div elements became actual buttons. Movie details expand in place, with aria-expanded tracking their state and aria-controls identifying the panel. Search submits a local form. Review, privacy and contact links describe their destinations.

```
<button type="button" aria-expanded="false"
        aria-controls="details-0" data-details="details-0"
        aria-label="View details: Echoes of Tomorrow">
  View Details
</button>
<section id="details-0" hidden>...</section>
```

3 / Image alternatives

The actual starter has five image elements. Four content photographs receive accurate descriptions, and the repeated featured photo is decorative, so it uses alt="". The photos are placeholders rather than movie posters.

```
<img src="../assets/movie1.jpg"
     alt="Dark grapes growing in clusters on a vine">
<img src="../assets/featured.jpg" alt="">
```

Step 1 | Step 2 | Step 3

Steps 4-5 / Contrast and forms

Readable text and usable feedback for invalid input.

4 / Five measured contrast corrections

Ratios below are calculated from the actual CSS foreground and background values. All five corrected pairs exceed the normal-text minimum of 4.5:1.

Text	Before	After
Navigation links	#ccc on #666 3.58:1	#fff on #333 12.63:1
Movie metadata	#999 on #f5f5f5 2.61:1	#555 on #f5f5f5 6.84:1
Featured text	#aaa on #eee 2.00:1	#555 on #eee 6.43:1
Newsletter text	#bbb on #fff 1.92:1	#555 on #fff 7.46:1
Footer text	#666 on #222 2.77:1	#ccc on #222 9.91:1

5 / Labels, errors and focus

Each input has a visible label whose for value matches its id. Newsletter inputs reference their error messages with aria-describedby. Validation sets aria-invalid, writes a useful error and focuses the first invalid field.

```
<label for="subscriber-email">Email address</label>
<input id="subscriber-email" type="email" required
       autocomplete="email" aria-invalid="false"
       aria-describedby="subscriber-email-error">
<p id="subscriber-email-error" class="field-error"></p>
```

Missing name: "Enter your name." Invalid email: "Enter a valid email address, such as name@example.com." Once corrected, the invalid state clears. A valid submission shows that the demo has been validated; it sends and stores no newsletter data.

Step 4: contrast | Step 5: forms

Steps 6-7 / Loading strategy

Reserve layout space and keep parsing independent of scripts.

6 / Image dimensions and lazy loading

All five images have width and height. The four movie thumbnails stay eager because they can be visible initially on a wide screen. The lower featured image uses loading="lazy" and decoding="async". This reserves the intended shape while deferring off-screen work.

```
<!-- Movie thumbnails may be visible immediately. -->
<img src="../assets/movie1.jpg" width="400" height="560"
     alt="Dark grapes growing in clusters on a vine"
     loading="eager" decoding="async">

<!-- The featured illustration is below the movie list. -->
<img src="../assets/featured.jpg" width="360" height="500"
     alt="" loading="lazy" decoding="async">
```

7 / External scripts with defer

The lab uses body-end placement plus defer for the two original external library scripts and the new lab app script. This follows this lab exercise; the separate E3 exercise demonstrates head-plus-defer. Both approaches can avoid parser-blocking downloads.

```
<!-- At the end of body: -->
<script src="https://cdn.jsdelivr.net/npm/jquery@3.7.1/dist/jquery.min.js"
        defer></script>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
        defer></script>
<script src="lab.js" defer></script>
```

What the comparison shows

The combined changes improved the measured local Performance score from 76 to 100 and Accessibility from 70 to 100. Individual changes were not benchmarked in isolation. Local-image origin and redirect changes are part of the comparison.

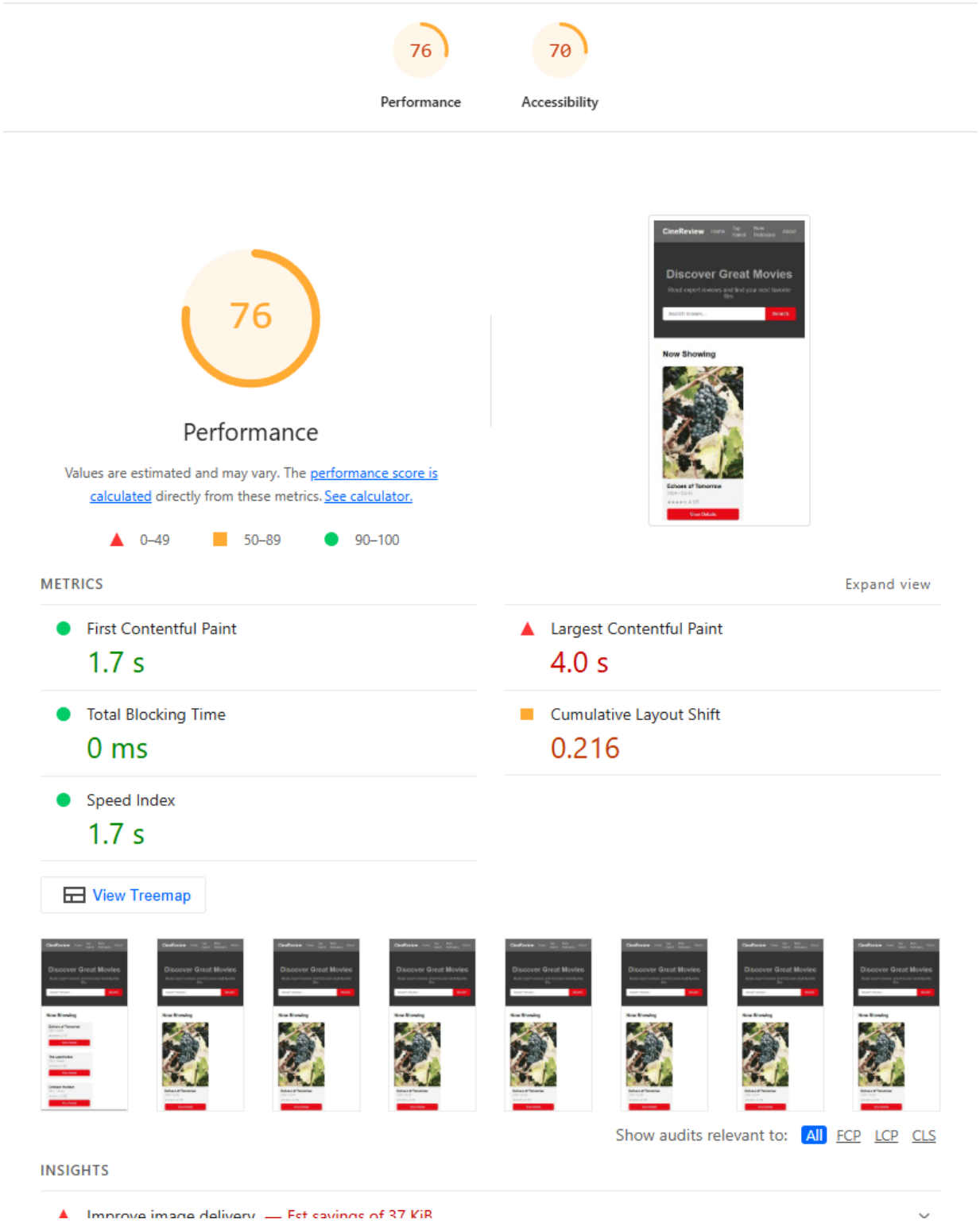
Step 6 | Step 7 | Completed solution

Source credits

CineReview starter: broken.html source. Placeholder photographs: Lorem Picsum. The retained baseline is unchanged; movie titles and review content are fictional.

Lab / Before: 76 Performance

Original saved Lighthouse report: score, metrics and page preview.

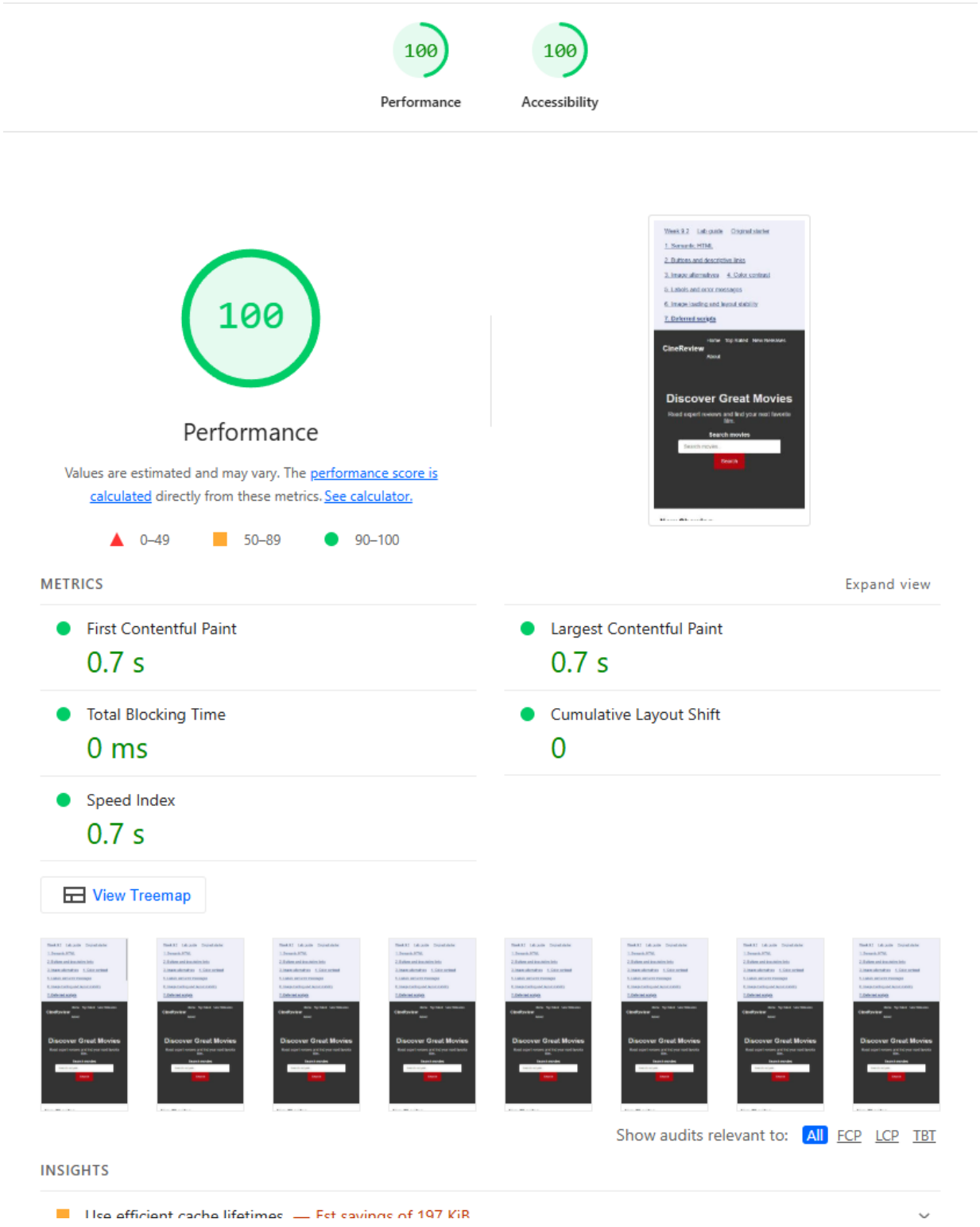


Audit URL: http://127.0.0.1:4173/week9_2/lab/broken.html
Captured: 2026-11-02T14:34:15.480Z (UTC).

Open the full HTML report | Open raw audit JSON

Lab / After: 100 Performance

Original saved Lighthouse report: score, metrics and page preview.



Audit URL: http://127.0.0.1:4173/week9_2/lab/solution.html
Captured: 2026-11-02T14:38:48.425Z (UTC).

Open the full HTML report | Open raw audit JSON